

Continuing Vannevar Bush's Vision in Science, *The Endless Frontier* (1945) Bush envisioned federal investment in basic research as an "endless frontier" driving progress, especially in the "war against disease" through sustained support for medical and biological inquiry. HHS's emerging role in accelerating AI adoption in clinical care extends this legacy—providing foundational infrastructure (data "libraries") and standards to enable private-sector innovation while upholding patient privacy, trust, and equity.

The government can best contribute by curating and maintaining large-scale, public-access repositories and standards for AI development/deployment, like how NIH/ONC already supports data ecosystems (e.g., via TEFCA's recent milestone of nearly 500 million records exchanged and draft USCDI v7 expansions, alongside federated approaches in recent strategic plans). This reduces barriers to entry for innovators, fosters competition, and aligns incentives for high-value AI tools.

Additionally, HHS can reduce data variability at the point of care by elevating standards for basic clinical documentation and measurement reliability—treating high-quality exam data as essential 'fuel' for AI models. Data is the food on which AI thrives, and improving the quality of medical exam data improves the fuel that AI models feed on.

Response to Question 1 (Barriers to Innovation and Adoption) A primary barrier is the fragmentation and inaccuracy of medical data, which limits robust AI training and reliable clinical deployment. Private innovators struggle with siloed, non-fungible datasets prone to errors (e.g., subjective measurements, incomplete records). This echoes pre-ImageNet challenges in computer vision, where lack of unified, large-scale labeled data stalled progress until ImageNet (launched ~2009 with 14M+ images) catalyzed breakthroughs like AlexNet (2012) and modern deep learning.

Response to Question 8 (Interoperability Opportunities) and Question 4 (Evaluation Methods/Support) HHS should prioritize:

1. **Expanded Medical Image Repository** — Develop or fund a unified, large-scale "ImageNet for Medicine" covering external body views, surgical/pathology/microscopy images, radiology (e.g., expanding on existing public resources like MIMIC-CXR [~377K chest X-rays with reports], CheXpert [~224K chest X-rays], NIH Chest X-ray [112K images], and others). This would include multimodal data (video, thermal, depth) for dynamic, touch-free exams, enabling better pre- and post-deployment evaluation, robustness testing (e.g., temporal sequences, physiological ratios), and benchmarking. HHS mechanisms like grants, cooperative agreements (e.g., CRADAs), or prize competitions could accelerate this, building on NIH/ONC interoperability initiatives (e.g., USCDI expansions, TEFCA for federated access).
2. **New Exam Data Standards for Fungibility and Reliability** — Establish standards for automated, low-error measurements (e.g., height/weight via sensors avoiding postural/food biases; vital signs via remote photoplethysmography or pressure floors). Incorporate time-series data and ratios (e.g., gait symmetry, respiratory effort over sequences) to reduce subjectivity and enable high-confidence AI decisions. This would expand market opportunities by making data interoperable across systems, fueling

research (e.g., longitudinal baselines for precision medicine), and supporting novel tools such as automated, touch-free physical exams (e.g., Exam OS concepts). HHS could drive this via ONC standards updates and NIH R&D investments.

These steps would lower barriers, promote trustworthy AI, reduce costs (e.g., commoditizing exams), and improve outcomes—directly advancing Bush's vision in the AI era. These investments would accelerate the translation of AI from concept to bedside, echoing Bush's call for federal support to unleash scientific progress in service of health.